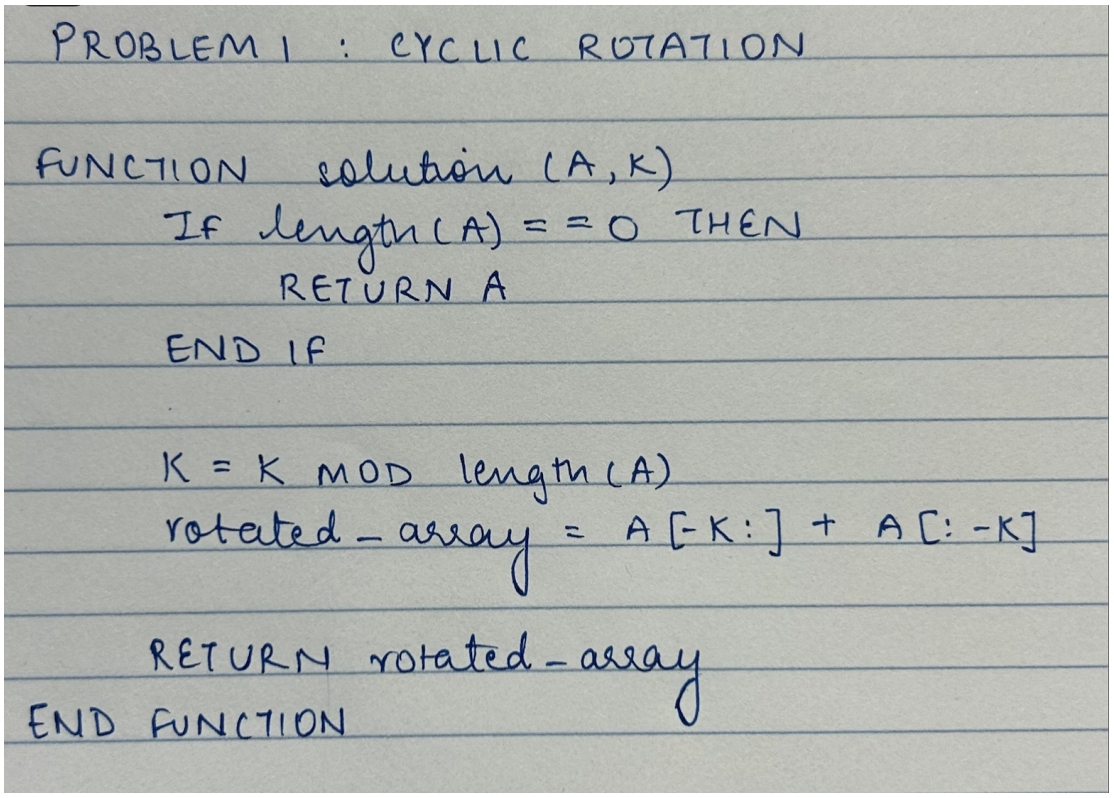
CISC 121
ASSIGNMENT-1

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#Problem 1- Cyclic Rotation

Pseudocode:



Code:

```
def solution(A, K):
    if len(A) == 0:  #Checking if the array is empty, if its empty return it as it is.
        return A

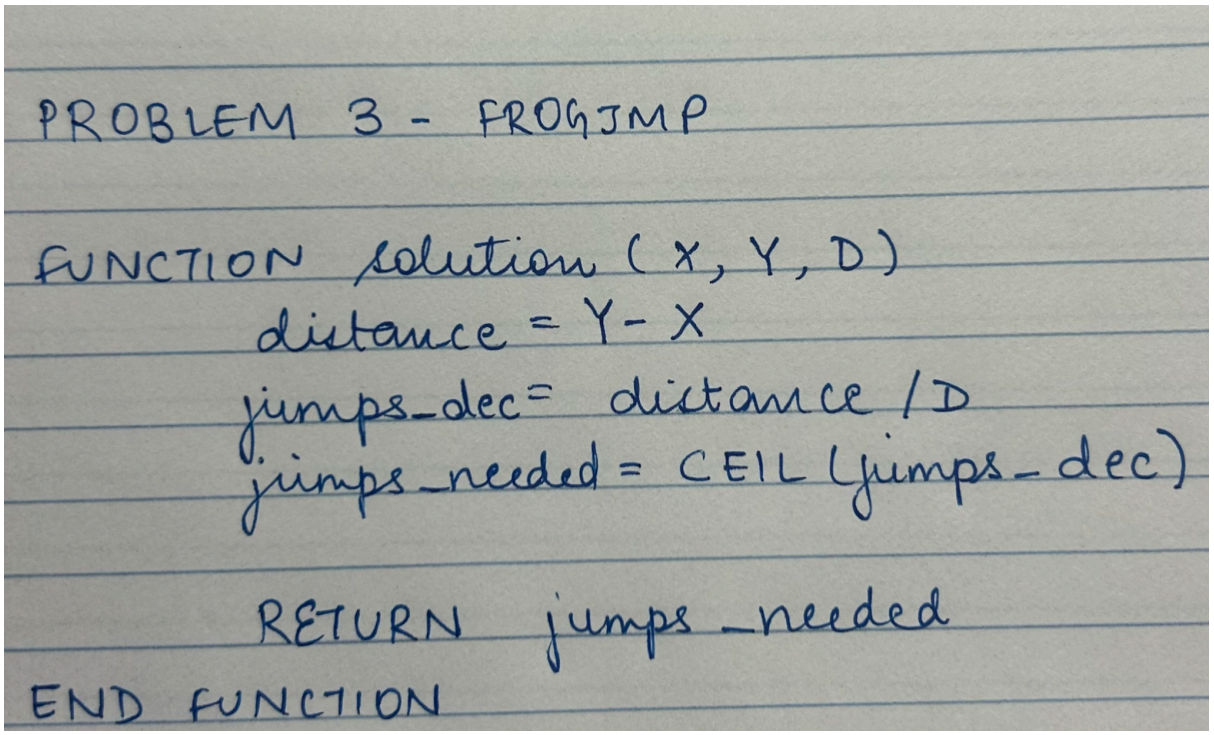
    #Calculating the effective number of rotations
    K = K % len(A)

    #Rotating the array to the right by K positions and storing it using return
    rotated_array = A[-K:] + A[:-K]

    return rotated_array
```

#Problem 3- FrogJmp

Pseudocode:



Code:

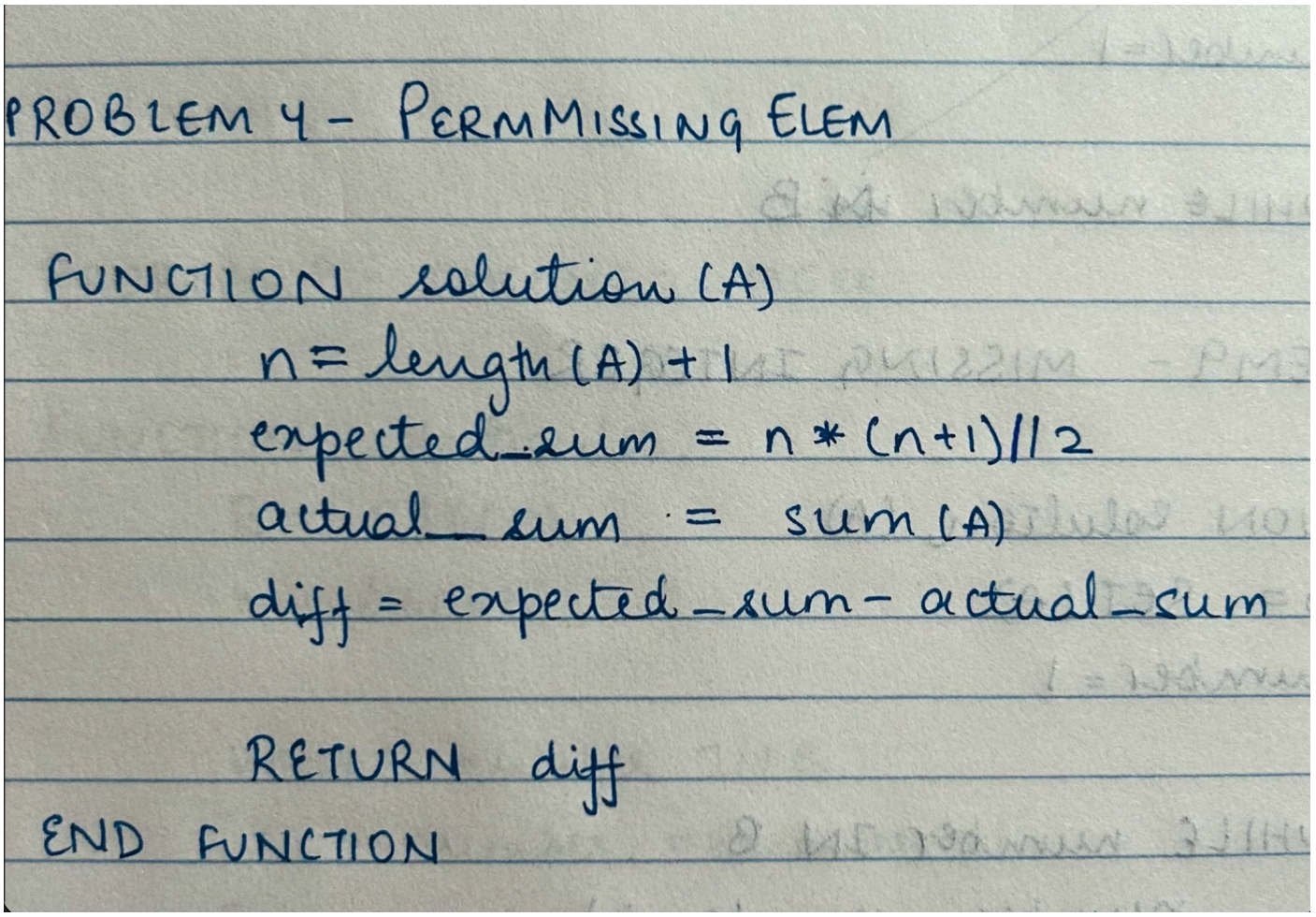
```python
import math
def solution(X,Y,D):
    #How far the frog still needs to travel
    distance= Y-X

    #Jumps needed by the frog (without roundup, possibly in decimals)
    jumps_dec= distance/ D
    #Round UP, because a partial jump still counts as a full jump
    jumps_needed= math.ceil(jumps_dec)

    return jumps_needed
```

#Problem4- PermMissingElem

Pseudocode:



Code:

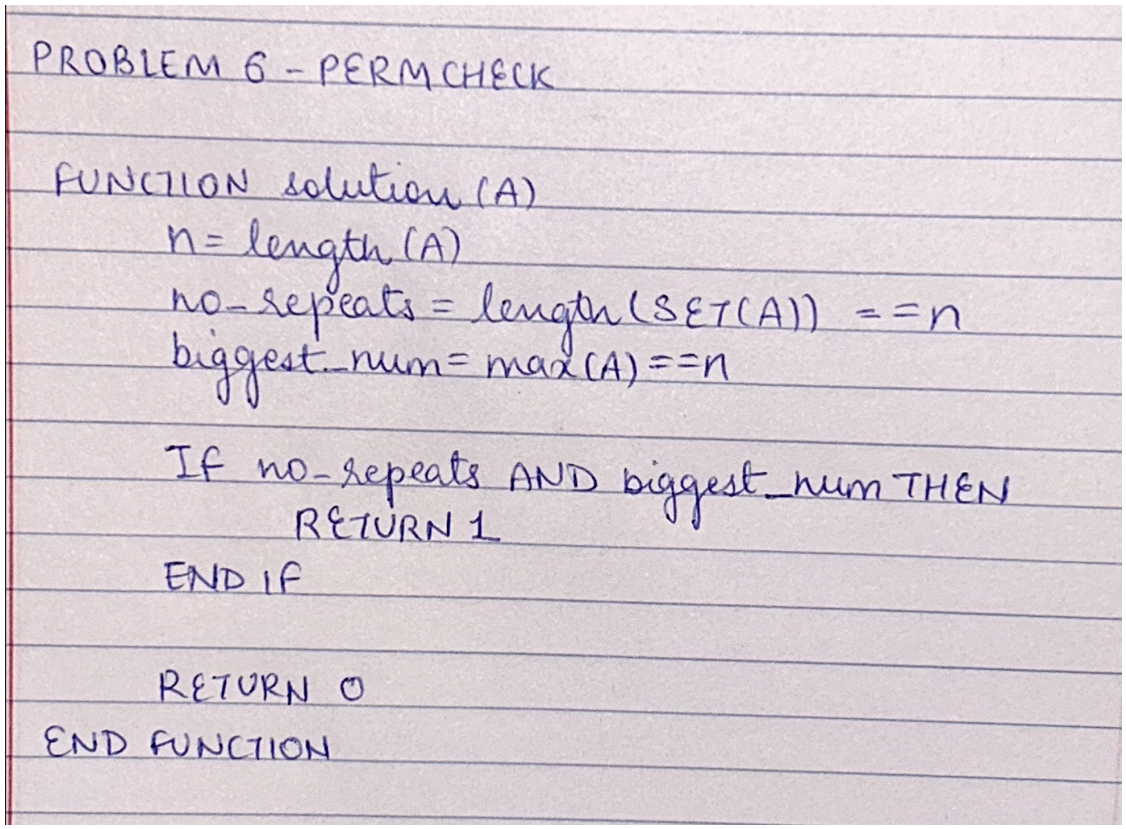
```
def solution(A):
    n = len(A) + 1
    #Array is missing one number, so full set is 1 longer than A

    expected_sum= n * (n + 1) // 2
    #Sum of the array if the missing element was present

    actual_sum= sum(A)
    #Sum of the entire array (excluding the one missing element)
    diff = expected_sum - actual_sum
    return diff
```

#Problem6- PermCheck

Pseudocode:



Code:

```
def solution(A):
    #Finding how many numbers are in the array
    n= len(A)
    #Converting the array into a set to remove duplicates
    #If the length stays the same, there were no repeated numbers
    no_repeats= len(set(A)) == n

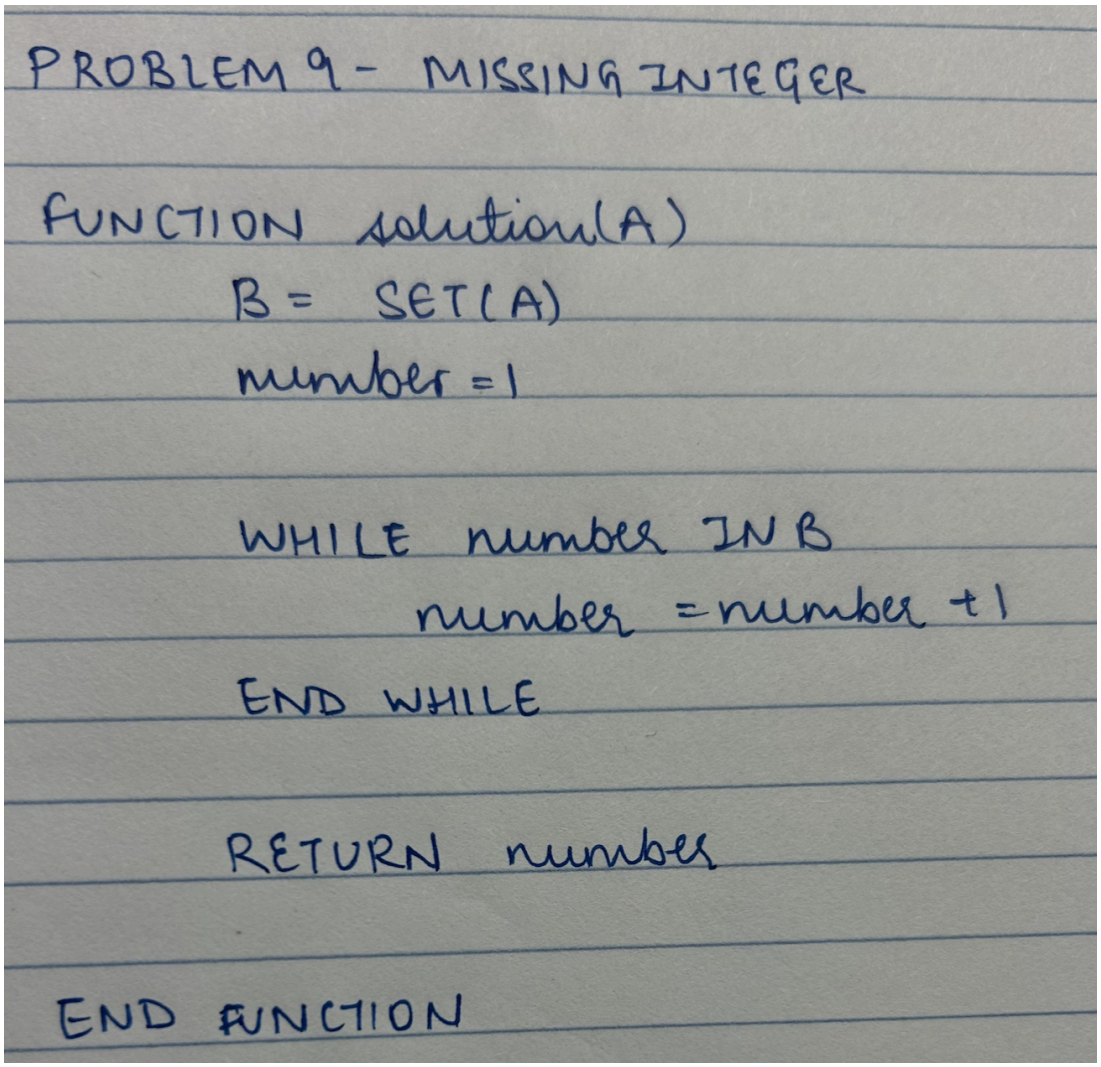
    #Checking if the largest number in the array is equal to n
    biggest_num= max(A) == n

    #If there are no duplicates and the biggest number is n, the array must contain
all numbers from 1 to n
    if no_repeats and biggest_num:
        return 1

    #Otherwise, the array is not a permutation
    return 0
```

#Problem9- MissingIntegers

Pseudocode:



Code:

```
def solution(A):
    #Converting the array into a set so we can remove duplicates
    B=set(A)
    #Start checking from 1, since we need the smallest positive integer
    number=1

    #We keep going as long as the current number is present in the set
    while number in B:
    #If it's there, move on and check the next integer
       number=number+1
    #The loop stopped, so this is the first positive integer that is missing
    return number
```

Citation:

- Open-source Resources:

  https://docs.python.org/3/library/math.html#math.ceil

- AI Usage

  I used Claude (Anthropic) to understand how // (floor/integer division) differs from / (true division) in Python in the context of the mathematical expression used in my code(in #Problem4), so that I could represent it correctly in my pseudocode. I wrote the pseudocode and code myself; the AI was used only to clarify this concept.

  Chat link:
  https://claude.ai/share/7640cfe0-092e-4a9a-9f6c-e25761b8f7ae

GitHub Repository Link:

https://github.com/bhumika1107/Assignment-1_CISC121.git